

Jashanpreet Singh

jashanpreetlehra@gmail.com | 7589011526 |  jashan645 |  Jashanpreet645

Summary

Results-driven Computer Engineering undergraduate with strong foundations in Data Structures, Algorithms, Object-Oriented Programming, and Full-Stack Software Development. Experienced in designing and deploying scalable web applications using the MERN stack, building RESTful APIs, and integrating AI/ML models into production-ready systems. Passionate about writing clean, maintainable code and following SDLC and Agile methodologies.

Technical Skills

- **Languages:** C++, Python, JavaScript, TypeScript
- **Core CS Concepts:** Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- **Frameworks & Libraries:** React, Node.js, Express.js, OpenCV, Tailwind, RESTful API Development
- **ML / AI:** Machine Learning, Deep Learning, NLP, Computer Vision, TensorFlow, PyTorch, Scikit-learn
- **Databases:** MongoDB, Oracle SQL, PL/SQL
- **Others:** Git, Github, Excel, Data Analysis, Problem Solving, Debugging, Technical Writing, JWT Authentication

Projects

Smart Athlete Risk Analyzer

Full Stack + AI/ML

- Developed a browser-based machine learning app to assess injury risk based on user-input physical metrics.
- Integrated TensorFlow.js for client-side inference, reducing latency by 60
- Built a modular, responsive UI with Next.js and Tailwind CSS, ensuring seamless performance across all devices.
- Created 10+ reusable React components for dashboards, input handling, data visualization, and real-time alerts.
- Enhanced user experience with client-side routing, lazy loading, and form validation, boosting engagement by 30

Krishi—Agricultural AI Platform

Full-Stack + AI/ML

- Built an end-to-end agricultural AI platform providing crop recommendation, fertilizer suggestion, and plant disease detection using ML and deep learning models
- Developed a React + TypeScript frontend with responsive UI and secure authentication, integrated with a Node.js + Express backend and MongoDB
- Deployed AI services on Hugging Face Spaces, including CatBoost-based crop recommendation and MobileNetV2-based plant disease detection via REST APIs
- Implemented JWT-based authentication, image upload pipelines, and API orchestration between backend

Work Experience

Research Intern, Thapar Institute of Engineering and Technology

1 June 2025 - 28 July 2025

- Worked Under Dr. Gurpal Singh Chhabra, Computer Science Department, TIET
- Conducting a comprehensive review on developed smart contracts using Solidity and integrated with Ethereum.
- Enhancing skills in scientific writing, critical analysis, and data interpretation, collaborating with domain experts throughout the research process

Education

B.E. Computer Engineering, Thapar Institute of Engineering and Technology

2023–2027

XII, D.A.V. Public School, Lehragaga

2023

Responsibilities

- Served as **Team Leader** in multiple hackathons (Leading development lifecycle from ideation to deployment)
- Volunteered in planning and management activities for **Thapar Food Festival, 2025**